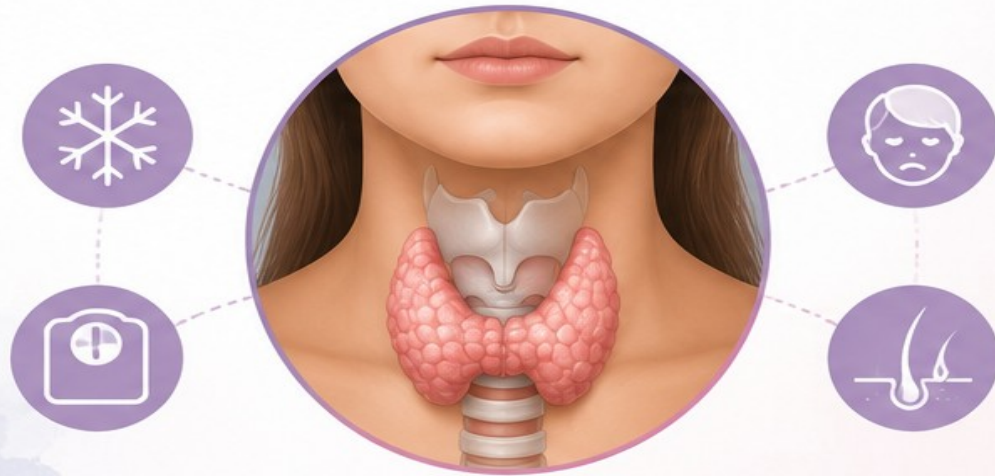


HYPOTHYROIDISM *in* PREGNANCY



Topic 4



Hypothyroidism and its significance during pregnancy

Hypothyroidism is decrease secretion of thyroid hormone

- **Primary maternal hypothyroidism:** presence of elevated Thyroid Stimulating Hormone (TSH) levels during pregnancy.
- **Overt hypothyroidism (OH):** Serum TSH (S. TSH) levels are elevated and Serum T4/Free T4 (FT4) levels are low. S. TSH ≥ 10 mIU/l is taken as OH irrespective of FT4 levels.
- **Subclinical (SCH):** S. TSH level is elevated (≤ 10 mIU/l) with normal Serum T4/FT4.
- **Positive thyroid antibody titers** suggest autoimmune thyroid disease.

Significance during pregnancy:

The foetus is dependent on maternal trans-placental thyroid hormone supply in the first trimester. To meet the increased demands, the thyroid hormone production increases by 50%.

Diagnostic equipment for Hypothyroidism

- Serum TSH level tested at facility to be collected
- if identified as HRP to be enlisted in PMSMA portal and intimated to ANM for HRP linelisting

Level of Facility	Diagnostic equipment
VHSND	11 Criteria for symptomatic screening
CHC Level ANC	Isolated TSH
PMSMA	Isolated TSH for all PW

Adverse effects of untreated hypothyroidism in mother and fetus/newborn

In mother

- Miscarriages
- Recurrent pregnancy losses
- Anaemia
- Pre-eclampsia/Eclampsia
- Gestational diabetes mellitus
- Abruptio placentae
- Postpartum hemorrhage
- Increased cesarean section due to fetal distress
- Rarely myopathy and
- Congestive heart failure in severe cases

In Fetus/newborn

- Preterm births
- Intrauterine growth retardation
- Intrauterine death
- Respiratory distress and consequent hospitalization
- Increased perinatal mortality
- Low birth weight
- Possibility of birth defects
- low IQ baby
- Cognitive, neurological and developmental impairment as thyroid hormone is critical for fetal brain growth

Magnitude of the problem and target population

Global: Overall 2-3%

India: Overall 4.8-12% with average of 10%

In India, screening for hypothyroidism is recommended for high risk pregnant women



HIGH RISK FACTORS FOR HYPOTHYROIDISM



Identify high risk women early for timely screening and better outcomes

1



Residing in an area of known moderate to severe iodine insufficiency (according to area mapping)

2



Obesity (pre-pregnancy/first trimester Body Mass Index (BMI) ≥ 30 kg/m²)

3



History of prior thyroid dysfunction or prior thyroid surgery

4



Symptoms of thyroid dysfunction or the presence of goiter

5



History of thyroid dysfunction in first degree relative (parents/siblings/children)

6



History of diagnosed mental retardation in family/previous births

7



Known case of autoimmune diseases like Type I diabetes/ Systemic Lupus

8



Erythematosus (SLE)/ Rheumatoid Arthritis (RA)/Addison's disease/ Coeliac disease, etc.

9



History of recurrent miscarriages, pre-term delivery, intrauterine demise, pre-eclampsia/ eclampsia, abruptio placentae

10



History of infertility (inability to conceive after one year of unprotected intercourse)

11



Use of amiodarone or lithium, or recent administration of iodinated radiologic contrast

Diagnostic criteria in pregnancy

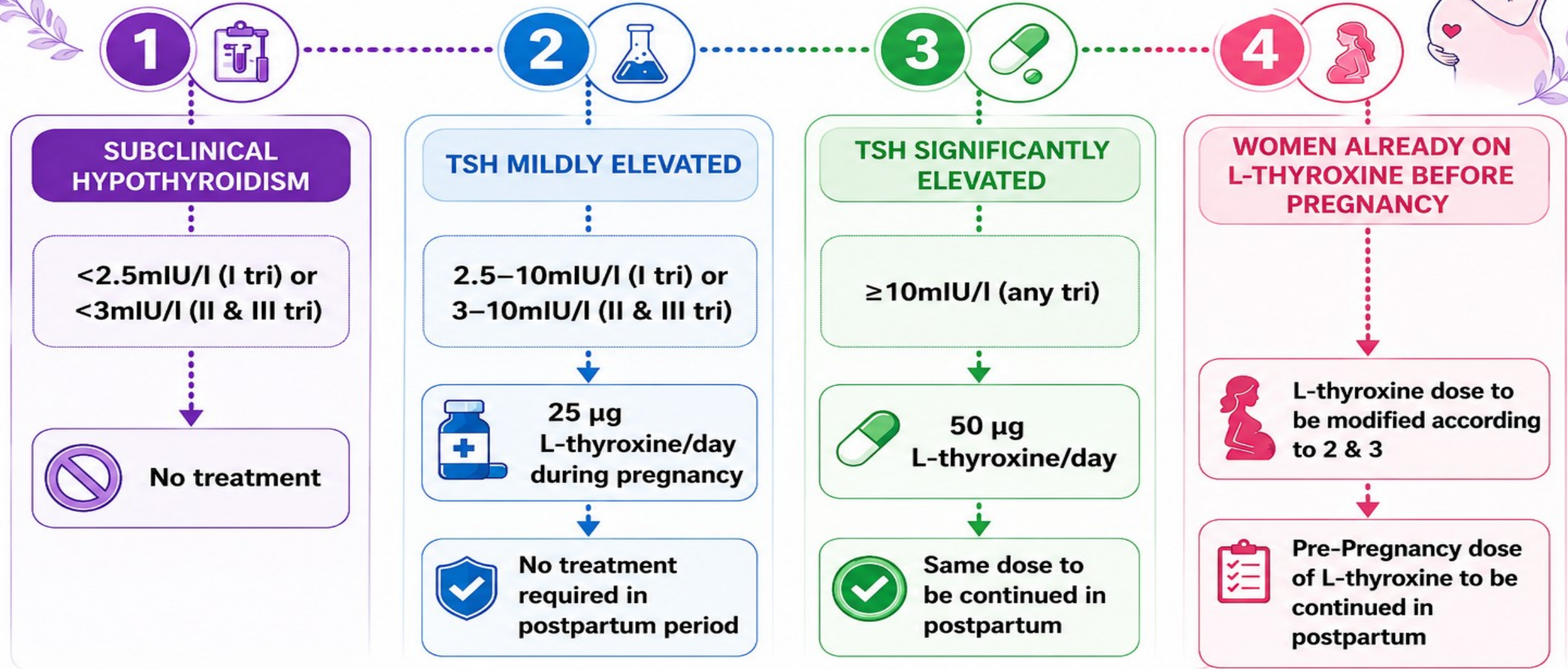
Normal range (reference levels for TSH) during pregnancy:

Trimester	Range (TSH)
First trimester	0.1-2.5 mIU/l
Second trimester	0.2-3 mIU/l
Third trimester	0.3-3mIU/l

Criteria for hypothyroidism during pregnancy

Type of hypothyroidism	Level of TSH mIU/L	Level of Free T4
Subclinical hypothyroidism	2.5 -10	Normal
Overt hypothyroidism	>2.5 - 3	Low
Overt hypothyroidism	>10	Any value (irrespective of T4)

FLOW CHART FOR TREATMENT OF HYPOTHYROIDISM IN PREGNANCY AND POSTPARTUM PERIOD



Repeat TSH 6 weeks postpartum and treat further as required

